PHYS 4041/5041
Fall 2023
Homework 4
Due: In class, Thursday, October 5, 2022

1. Consider the general linear second-order ODE

$$p_0 y'' + p_1 y' + p_2 y = -\tilde{\lambda} y, \tag{1}$$

where p_0, p_1, and p_2 are functions of x and $\tilde{\lambda}$ is a constant parameter. The standard form of the Sturm-Liouville ODE

$$(py')' + qy = -\lambda w y, \tag{2}$$

where p, q, and w are functions of x and λ is a constant parameter.

(a) Assuming that p_0, p_1, p_2, and $\tilde{\lambda}$ are given, multiply Eq. 1 by w. By comparing to Eq. 2, show that the unknown p, q, w, λ must satisfy $p = wp_0$, $p' = wp_1$, $q = wp_2$, and $\lambda = \tilde{\lambda}$.
(b) Show that $p = \exp(\int \frac{p_1}{p_0} dx)$.
(c) Show that $w = \frac{1}{p_0} \exp(\int \frac{p_1}{p_0} dx)$.

This shows that a general linear second-order ODE can, in principle, be put into the standard form of the Sturm-Liouville ODE.

2. For each of the ODEs below, find the weight function $w(x)$ that will allow the ODE to be written as a standard Sturm-Liouville ODE. Also give the resulting ODE in the standard form.

(a) $y'' + (1 - x)y' + \lambda y = 0$ (Laguerre ODE)
(b) $y'' - 2xy' + 2\lambda y = 0$ (Hermite ODE)
(c) $x^2 y'' + xy' + (x^2 - \lambda^2)y = 0$ (Bessel ODE)

3. Consider the Sturm-Liouville ODE $y'' = -\lambda y$ over the interval $[0, 1]$. The ODE is solved by solutions of the form $y = A \exp(\gamma x) + B \exp(-\gamma x)$, where $-\lambda = \gamma^2$ and γ is real.

(a) For the Dirichlet boundary conditions $y(0) = y(1) = 0$, show that that it is impossible for solutions of this form to match the boundary conditions.
(b) For the Neumann boundary conditions $y'(0) = y'(1) = 0$, show that that it is impossible for solutions of this form to match the boundary conditions.

4. The set of functions $\left\{u_n = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}\right\}$, with $n = 1, 2, 3, \ldots$ is orthonormal and complete over $(0, L)$.

(a) Express the function $f(x) = 1$ in this basis.
(b) Express the function $f(x) = x$ in this basis.

5. Here we will show a remarkable property of the Sturm-Liouville Green Function for homogeneous Dirichlet boundary conditions. Dirichlet boundary conditions describe the situation when the value of the function is specified on the endpoints (as opposed to the derivative).

(a) For (almost) any p, q, y_1, and y_2 which are real-valued functions of x, use integration by parts to show that

$$\int_a^b y_1[(py_2')' + qy_2]\,dx - \int_a^b y_2[(py_1')' + qy_1]\,dx = p(y_1y_2' - y_2y_1')\Big|_a^b,$$

where $b > a$ are real constants.

(b) Suppose that we have somehow found the Green Function $G(x, x')$ solving

$$(pG')' + qG = \delta(x - x')$$

subject to $G(a, x') = G(b, x') = G(x, a) = G(x, b) = 0$, i.e., homogeneous Dirichlet boundary conditions. Now consider the following inhomogeneous ODE:

$$(py')' + qy = f,$$

where f is a real function of x, and y is subject to *arbitrary* Dirichlet boundary conditions, i.e., $y(a)$ and $y(b)$ are specified but may be non-zero. By using $y_1 = y$ and $y_2 = G(x, x')$ in formula derived in part (a), show that the solution to this differential equation is given by

$$\begin{aligned} y(x') &= \int_a^b G(x, x')\,f(x)\,dx + py\frac{\partial G}{\partial x}\Big|_{x=a}^{x=b} \\ &= \int_a^b G(x, x')\,f(x)\,dx + p(b)y(b)\frac{\partial G}{\partial x}(b, x') - p(a)y(a)\frac{\partial G}{\partial x}(a, x'). \end{aligned}$$

Remarkably, the Green Function for homogeneous boundary conditions also can be used to find the solution for inhomogeneous boundary conditions.